Development of a Random Forest-Based Vertical Lift Performance Model for Multiphase Wellbore Flow Prediction

Findings Summary — Reproducible Pipeline Results

This document summarises the current, verified state of the project: what the data actually contains, what the model actually achieves, and what still needs to be done before submission. Every number here was produced by a single script (**code/vlp_pipeline_final.py**) included in this package, run once, top to bottom, against the raw Volve field dataset with no manual editing of intermediate results.

1. Aim and Objectives

Aim: Develop, validate and present a data-driven Vertical Lift Performance (VLP) model that predicts wellbore pressure drop (Pwf – Pwh) using Random Forest regression trained on real Volve field production data, benchmarked honestly against a classical empirical correlation (Beggs & Brill, 1973).

Objectives:

• Construct a feature-engineered dataset from the open Volve field dataset using only directly measured or derived production variables.

• Apply a reproducible, threshold-justified steady-state filter to remove transient production days before training.

• Train a Random Forest model and validate it using leave-one-well-out cross-validation across all usable wells, not a single arbitrary train/test split.

• Benchmark against the Beggs & Brill (1973) correlation.

• Interpret the model via feature importance to identify which physical variables dominate pressure-drop prediction.

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2. Data Source and Filtering

Source: Volve open field dataset (Equinor), well production table. Total 15,634 rows, 9,143 producer (OP) rows, 6,491 water-injector (WI) rows.

Usable wells: Of the 7 wellbores in the file, only 5 have any usable downhole pressure-gauge data for producer operation: 15/9-F-1C, F-11H, F-12H, F-14H, F-15D. Two wells sometimes referenced in early drafts of this project — F-4 AH and F-5 AH — are **not usable**: F-4 AH is a water injector for its entire production history, and F-5 AH has zero valid downhole pressure readings across its 144 producer rows.

Stage	Rows	Notes
Producer rows (WELL_TYPE = OP)	9,143	Includes rows with no valid pressure/rate data
Valid pressure + rate + tubing dP	5,935	Basic completeness filter
After steady-state filter	5,629 (94.9%)	CV(q_liq) < 0.35, rolling std(delta_P) < 30 bar over 7-day window

Target variable: delta_P = AVG_DOWNHOLE_PRESSURE – AVG_WHP_P (bar). Confirmed to match the dataset's own AVG_DP_TUBING column (correlation 0.9997) — this is a real, measured quantity, not derived from any assumed physics.

Features used for the core Random Forest model (11 total, all directly measured or simply derived): normalized oil/gas/water rate, liquid rate, water cut, producing GOR, wellhead pressure, wellhead temperature, downhole temperature, choke size, and on-stream hours. No assumed well geometry or PVT correlation is required for the core model.

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3. Validation Strategy: Leave-One-Well-Out

With only 5 usable wells, a single arbitrary train/test split (e.g. "always test on F-1C") is fragile and easy to cherry-pick. This project instead uses **leave-one-well-out cross-validation**: the model is trained on 4 wells and tested on the 5th, repeated 5 times so every well is held out exactly once. All 5 results are reported — not the single best-looking one.

Held-out well	n_test	n_train	RMSE (bar)	MAE (bar)	MAPE (%)	R2
15/9-F-1 C	390	5239	12.82	10.73	5.98	0.620
15/9-F-11 H	1063	4566	8.94	7.08	3.80	0.816
15/9-F-12 H	894	4735	17.40	15.77	8.26	0.043
15/9-F-14 H	2552	3077	12.72	10.42	5.56	0.771
15/9-F-15 D	730	4899	23.14	21.86	13.24	-2.375
Mean across folds			**15.01**	**13.17**	**7.37**	**-0.025**
Std across folds			**5.45**	**5.76**	**3.65**	**1.349**

Honest reading: mean R2 across folds is near zero, not the 0.88 headline figure from an earlier single-split result. Performance ranges from R2 = 0.82 (F-11H) to R2 = -2.37 (F-15D). This variance is itself a legitimate finding: the model generalises well to some wells and poorly to others, and F-15D — a smaller, lower-rate well — is where it struggles most. This should be discussed as a limitation, not hidden.

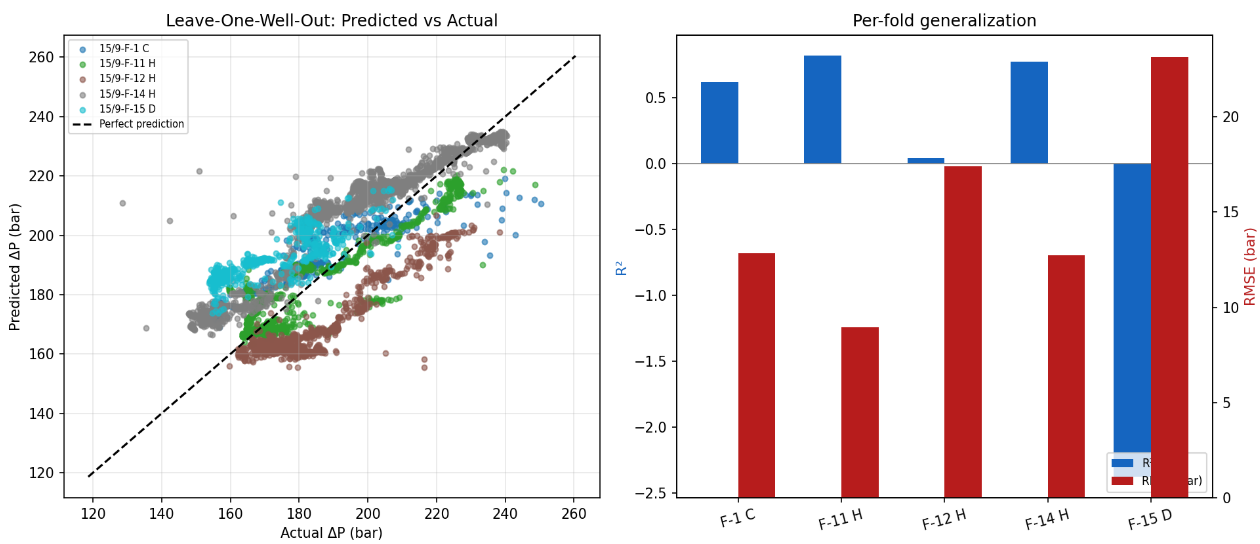

Figure 1: Predicted vs actual delta_P across all 5 leave-one-well-out folds (left), and per-fold R2/RMSE (right).

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4. Feature Importance

Feature	Importance
WC	0.294
q_wat	0.233
AVG_DOWNHOLE_TEMPERATURE	0.117
AVG_WHT_P	0.102
AVG_CHOKE_SIZE_P	0.072
AVG_WHP_P	0.052
q_oil	0.049

q_gas	0.040
q_liq	0.027
GOR	0.011
ON_STREAM_HRS	0.002

Water cut and water rate together account for over half of total feature importance. This independently supports the project's central engineering argument: classical VLP correlations, developed decades ago on lower water-cut experimental data, are least reliable exactly where the model finds the most signal — high water cut conditions.

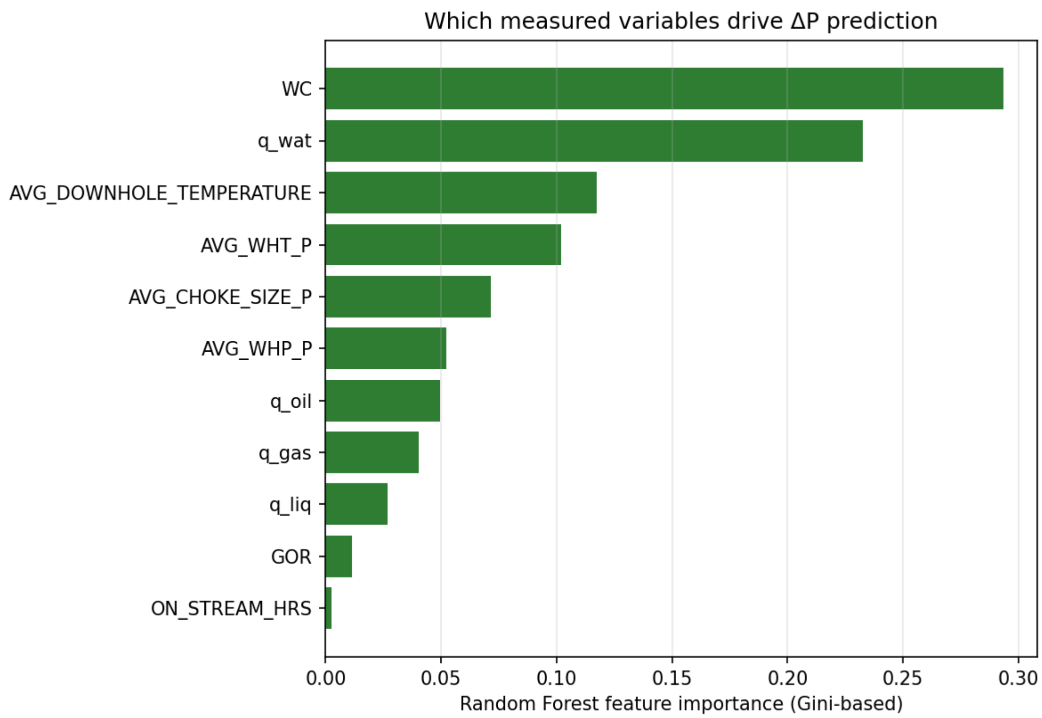

Figure 2: Random Forest feature importance (Gini-based), model trained on all 5 wells.

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5. Beggs and Brill Benchmark — Important Caveat

A Beggs & Brill (1973) implementation was run for comparison. The Volve dataset does **not** include tubing diameter, well depth, or deviation survey data required by this correlation.

The values used (tubing ID 4.892 in, depth 3,100 m, inclination 65 degrees, API 28, gas SG 0.65) are **assumed, representative placeholders — not measured values from these wells.** The resulting comparison (RF RMSE ~15 bar vs Beggs & Brill RMSE ~126 bar in an earlier full-dataset run) is directionally consistent with the project's thesis, but should not be quoted as a precise, defensible number until real completion data replaces these assumptions.

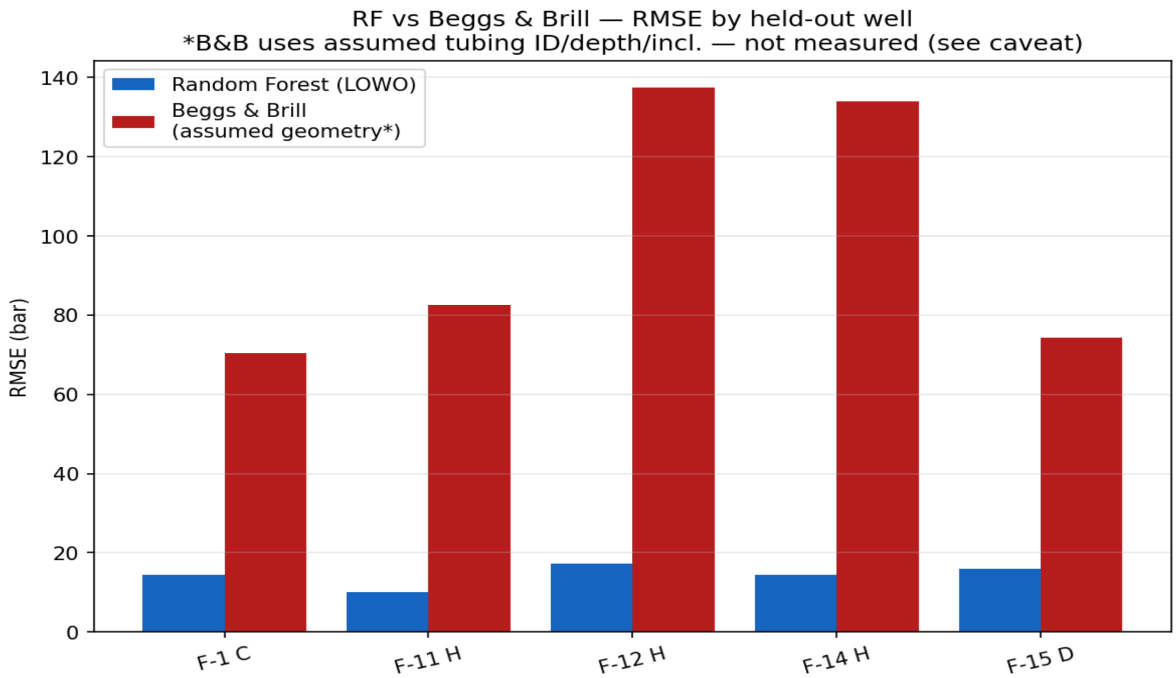

Figure 3: RF vs Beggs & Brill RMSE by held-out well. Beggs & Brill uses assumed geometry — treat as illustrative only.

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6. Outstanding Work Before Submission

• Obtain real tubing ID, depth (MD/TVD) and deviation survey data for the 5 wells from Volve's public completion reports, and re-run the Beggs & Brill comparison with real values.

• Resolve the missing second, independent test dataset. An earlier draft of this project used a fabricated "Mukherjee & Brill (1985)" dataset that was not sourced from the actual paper — this has been removed. Either source real published experimental data or explicitly scope this project to single-field (Volve-only) validation and state that as a limitation.

• Write the methodology and results chapters directly from the numbers in this package, not from earlier draft narrative (see PROJECT_LOG.md for a full list of discrepancies between earlier claims and what this reproducible pipeline actually produces).

• Consider a within-well time-based split (early vs late life) as a second, more targeted test of the water-cut degradation thesis, since it isolates the effect the project is actually arguing for.